

Current internship goal: optimize guest programs for zkVM

April Goal

*take a look at combination of
guest program zkVM:*

ethrex ↔ ZisK

Method

*measuring AIR-costs of
mainnet block execution*

*ZisK profiling is super helpful
and provides top cost function
calls*

Result

*3 PRs of total speed-up on
eip-8025 execution path:*

up to 26%

Current changes

#6549 Merged

measured speed-up: ~21%

use hash precompiles in eip-8025 hash tree root
eip-8025 computation path was bypassing precompiles

#6560 In review

measured speed-up: ~4.5%

remove rayon from eip-8025 execution path
parallelism is not applied in zkVM, rayon is pure overhead

#6588 In review

measured speed-up: ~0.5%

remove sync from eip-8025 execution path
locks are not required in zkVM, sync is pure overhead

Next steps

01 Cover remaining cases for unused hash precompiles

Ratio of the total AIR costs: ~3%

02 Return to rewriting `rkyv` serialization of `ExecutionWitness` to SSZ

Ratio of the total AIR costs: ~11%

03 Check if we can use native zisk 256-bit arithmetic on `U256::div_mod`

Ratio of the total AIR costs: ~5%

If you have any suggestions feel free reach out!